



Grade K Mathematics

OPTIMA ACADEMY ONLINE · FAMILY COURSE OVERVIEW · KINDERGARTEN · FULL YEAR

~12½ hrs per week · 150 min/day, M–F

About this overview. This document is an example of how the course might be taught. This overview is drawn from the Florida course description and Optima's 2026–27 course plan rather than a fully built course, so scholars may see some variation in lesson order, activities, and examples from course to course, and between the live school and On-Demand. Each teacher also leans into their own strengths and passions; the structure, expectations, and time commitment below are representative of the course as designed.

Grade K Mathematics introduces Kindergarten scholars to essential mathematical concepts including counting and cardinality, the relationship between operations and numbers, comparison and classification of attributes, and recognition of two- and three-dimensional shapes, enriched with classical pedagogy and critical thinking. Through engaging, inquiry-based experiences, scholars develop early numeracy and problem-solving skills. Following Optima's Singapore-style concrete-pictorial-abstract approach, each new idea is met first with hands-on models, then pictures, then numbers.

THE YEAR AT A GLANCE

1 Counting and Cardinality (area of emphasis)

Counting to represent the total number of objects in a set and to order objects within a set, building one-to-one correspondence and number sense.

3 Measuring, Comparing, and Shapes (area of emphasis)

Measuring, comparing, and categorizing objects by various attributes, including their two- and three-dimensional shapes.

2 Addition, Subtraction, and Counting (area of emphasis)

Developing an understanding of addition and subtraction and the relationship of these operations to counting, using objects, drawings, and equations.

4 Mathematical Thinking and Reasoning (strand)

Reasoning through tasks, representing ideas in more than one way, and explaining thinking with objects and drawings, woven through the whole year.

A TYPICAL WEEK

- A new lesson most days, taught concrete-pictorial-abstract: hands-on models first, then pictures, then numbers.
- Daily counting practice and number-sense routines that build one-to-one correspondence and cardinality.
- Hands-on work with counters and everyday objects to model adding, taking apart, comparing, and sorting.
- Regular check-ins as topics close to see what each scholar has mastered.

TIME COMMITMENT

750 minutes / week

≈ 2½ hours a day, Monday–Friday

- The daily lesson and its hands-on practice fill most of the scheduled math time.
- Short counting and number-sense routines open or close many lessons.

WHAT SCHOLARS NEED

- A math notebook, pencil, and paper.
- Simple counters or everyday objects for the hands-on step of new concepts.
- A computer with reliable internet and access to Canvas.
- An adult to support young learners (especially in Kindergarten).

On-Demand (asynchronous). Scholars move through the course at their own pace, with due dates to keep the year on track. An Optima teacher is available throughout for support, feedback, and grading.

Live school. Live-school scholars take this same course on Optima's VR campus with a teacher and classmates, meeting daily as a core course.